PES University — Department of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #40

Student Startup Crowdfunding Platform

Retail, E-Commerce & Finance

Target Stakeholders / Actors	Student Founder, Campaign Backer (supporting: Faculty Committee, Payment Gateway)
Deliverables Included	1. Requirements Table (FR-001–FR-005, NFR-001–NFR-002) 2. UML Use-Case Diagram (with «include» and «extend») 3. Use-Case Flow Specification
Submission Format	Single consolidated PDF, submitted via GitHub repository

1. Requirements Specification

The requirements below are derived from the problem context: a crowdfunding engine for university student ventures that supports milestone-based funding targets, backer reward-tier selection, and verified fund releases upon Faculty Committee review.

1.1 Functional Requirements

ID	Requirement / Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow a Student Founder to create a crowdfunding campaign by defining an overall funding target broken into sequential milestones, each with a description, target amount, and expected completion date.	High	Pass: Campaign is saved with at least one milestone and the sum of milestone targets equals the overall funding goal. Fail: Campaign is saved with zero milestones, or milestone totals do not reconcile with the overall goal.	Milestone-based funding is the platform's core differentiator and is explicitly required by the problem statement.
FR-002	The system shall allow a Campaign Backer to browse an active campaign, select an available reward tier, and pledge an amount consistent with that tier's minimum contribution.	High	Pass: Pledge is recorded against the chosen tier and the backer receives a confirmation. Fail: A pledge is accepted below the tier's minimum, or against a tier with no remaining inventory.	Backer reward-tier selection is explicitly listed as a required stakeholder capability.
FR-003	The system shall process backer pledge payments through a PCI-DSS compliant payment gateway using tokenized authorization, holding pledged funds in escrow until the corresponding milestone is verified.	High	Pass: A payment token is generated, funds are held in an escrow ledger, and no raw card data is persisted. Fail: Card data is stored in plaintext, or funds are released before an escrow hold is confirmed.	Directly derived from the NFR-001 guideline; required to make FR-002 pledges financially actionable and secure.
FR-004	The system shall allow a Student Founder to submit evidence (documents / progress reports) against a specific milestone for Faculty Committee review, transitioning that milestone's status to “Pending Review”.	High	Pass: Evidence is attached and the milestone state transitions to Pending Review. Fail: The milestone state changes with no attached evidence, or evidence is accepted after the campaign has closed.	Faculty Committee review of milestone evidence is explicitly required before any funds are released.
FR-005	The system shall allow the Faculty Committee to review submitted milestone evidence and, upon approval, trigger disbursement of the corresponding funding tranche to the Student Founder; upon rejection, the milestone shall revert to “In Progress” with reviewer comments attached.	High	Pass: An approved milestone releases exactly the allocated tranche amount to the founder's payout account and logs the reviewer ID and timestamp. Fail: Funds are disbursed without an approval record, or the disbursed amount does not match the milestone's allocated amount.	Given as the sample requirement in the problem statement; this is the platform's core trust mechanism (verified, milestone-gated fund release).

1.2 Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The payment processing gateway must comply with PCI-DSS guidelines with tokenized payment authorizations, and confirm authorization within acceptable latency under peak load.	High	Pass: Benchmark tests confirm p95 authorization latency under 2 seconds at 500 simulated concurrent transactions, and a security scan finds no stored PAN data. Fail: Latency exceeds the threshold, or raw card data is found in logs or the database.	Given as the sample NFR in the problem statement; mandatory for any platform that moves real money on behalf of student ventures and backers.
NFR-002	Availability & Reliability	The platform shall remain available during active campaign funding windows and must not lose pledge or milestone-evidence data.	Medium	Pass: Uptime monitoring confirms ≥99.5% availability over a rolling 30-day window, and disaster-recovery drills show zero data loss. Fail: Uptime falls below the threshold, or a recovery drill results in lost pledge or milestone data.	Campaigns and milestone reviews are time-bound; downtime could cause backers to miss pledge deadlines or founders to miss evidence-submission windows, undermining platform trust.

2. UML Use-Case Diagram

The diagram models the system boundary of the Student Startup Crowdfunding Platform, its two primary stakeholder actors (Student Founder, Campaign Backer) and two supporting actors (Faculty Committee, Payment Gateway), and the primary use cases derived from the requirements above. It includes one «include» relationship (Pledge to Campaign mandatorily includes Authorize Payment and Select Reward Tier) and one «extend» relationship (Release Fund Tranche conditionally extends Review Milestone Evidence, only when evidence is approved; Request Refund conditionally extends Pledge to Campaign, only when a funding deadline is missed).

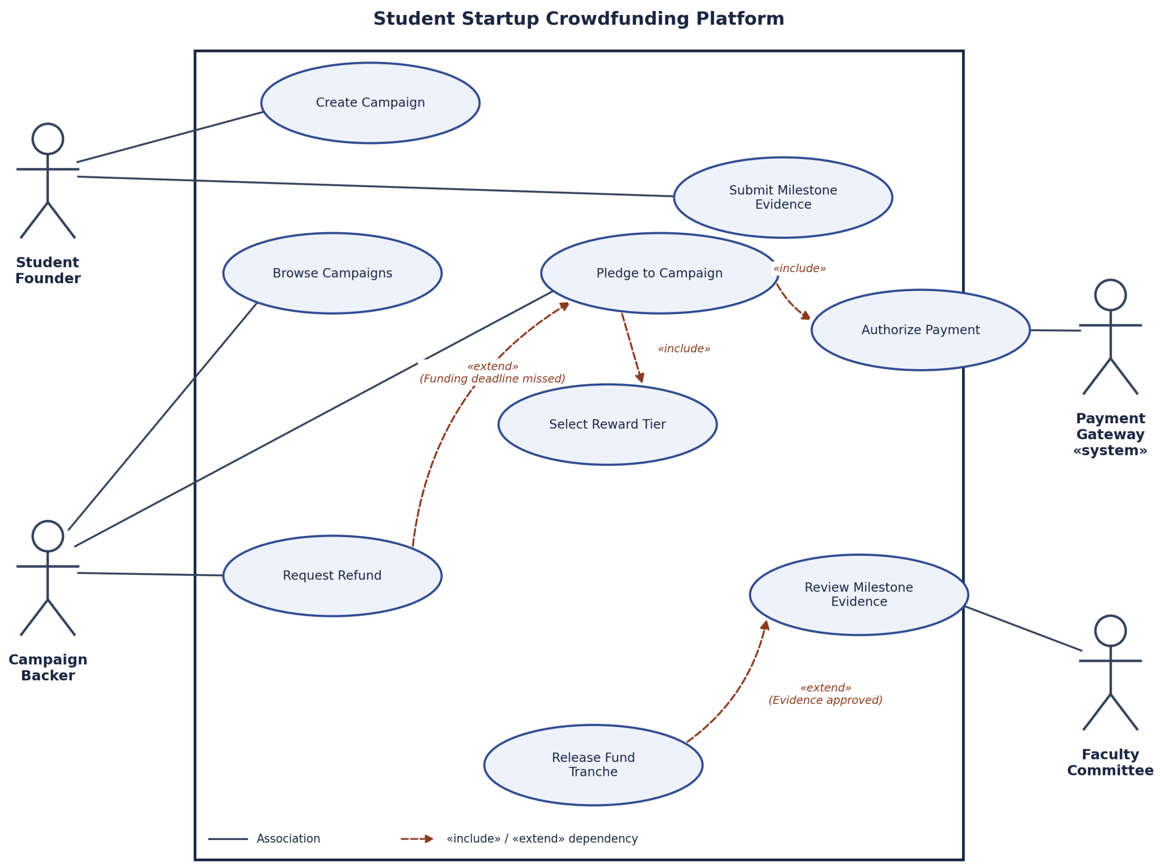

Figure 1: UML Use-Case Diagram — Student Startup Crowdfunding Platform

3. Use-Case Flow Specification

Use Case: Verify Milestone Evidence and Release Fund Tranche

Use Case ID	UC-05 (realizes FR-004 and FR-005)
Primary Actor	Faculty Committee
Secondary Actors	Student Founder, Payment Gateway
Trigger	Student Founder submits milestone evidence, placing a milestone into “Pending Review” status.

Preconditions

- The campaign is active and has at least one milestone in “Pending Review” status, with evidence attached by the Student Founder.
- The Faculty Committee member is authenticated and holds reviewer privileges for the campaign's department.
- Backer funds for the corresponding tranche are held in an escrow balance via the Payment Gateway.

Postconditions

- **Success:** Milestone status = “Approved”; tranche funds are transferred from escrow to the Student Founder's payout account; the transaction is logged with reviewer ID and timestamp; the Student Founder and all Backers are notified.
- **Failure / Rejection:** Milestone status = “Revision Requested”; escrow funds remain held; reviewer comments are recorded; the Student Founder is notified.

Main Success Scenario

1. The Faculty Committee selects a milestone with status “Pending Review” from the review queue.
2. The system displays the milestone's evidence, target tranche amount, and campaign details.
3. The Faculty Committee evaluates the evidence against the milestone's acceptance criteria.
4. The Faculty Committee submits an “Approve” decision, with optional comments.
5. The system validates that the requested tranche amount does not exceed the remaining escrow balance for the campaign.
6. The system requests a fund transfer from the Payment Gateway (escrow → founder payout account).
7. The Payment Gateway confirms the transfer and returns a transaction reference.
8. The system updates the milestone status to “Approved” and logs the reviewer ID, timestamp, and transaction reference.
9. The system notifies the Student Founder and all Backers of the milestone approval and fund release.

Alternate Flow A1 — Evidence Rejected

- **4a.** At step 4, the Faculty Committee submits a “Reject” decision with mandatory comments explaining the gap.
- **5a.** The system updates the milestone status to “Revision Requested” and does not contact the Payment Gateway.
- **6a.** The system notifies the Student Founder with the reviewer's comments and reopens the milestone for a new evidence submission.
- **7a.** The use case ends without fund release; escrow funds remain held pending resubmission.